

# VALENTINA ZANGIROLAMI

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**Research Interests** Statistical reinforcement learning, Deep learning, Statistical learning, Safe sequential decision-making.

**Research Network** Member of Prof. **Chengchun Shi**'s research group, contributing to research discussions on reinforcement learning, large language models, and reinforcement learning from human feedback.

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## Education

**Visiting PhD Student in Statistics** Feb 2025 – July 2026

London School of Economics and Political Science London, UK

- Research interests: *Statistical inference for reinforcement learning with temporally correlated rewards.*
- **Supervisor:** Prof. Chengchun Shi

**PhD in Statistics** Oct 2022 – Oct 2026

University of Milano-Bicocca Milan, Italy

- Research interests: *Statistical reinforcement learning with partial observability.*
- **Supervisor:** Prof. Chengchun Shi **Co-supervisor:** Prof. Matteo Borrotti
- Expected graduation at February 2027 (under italian timeline).

**Master's Degree in Statistical and Economic Sciences** Oct 2019 – Jan 2022

University of Milano-Bicocca Milan, Italy

- Thesis: *Recurrent structures of deep reinforcement learning for self-driving cars.*
- **Supervisor:** Prof. Matteo Borrotti **Co-supervisor:** Prof. Antonio Candelieri
- Grade: 110/110 with honors

**Bachelor's Degree in Statistical and Economic Sciences** Oct 2016 – July 2019

University of Milano-Bicocca Milan, Italy

- Thesis: *The mobile telecommunications market: the entry of Iliad.*
- **Supervisor:** Dott. Christian Garavaglia

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## Publications

### Refereed Journal Articles

- **Zangirolami, V.** and Borrotti, M. (2024). *Dealing with uncertainty: Balancing exploration and exploitation in deep recurrent reinforcement learning.* *Knowledge-Based Systems*, 293.

### Book Chapters and Proceedings

- **Zangirolami, V.**, Pavesi, F., and Zanotti, M. (2026). *Safe exogenous-state reinforcement learning for water tank systems.* *Statistical Science: From Theory to Applied Research IV*, accepted (to appear).
- Pedrazzini, E., **Zangirolami, V.**, Migliorati, S., and Borrotti, M. (2026). *Beta-Distributed Proximal Policy Optimization for Autonomous Driving with Grad-CAM Explainability.* *Artificial Intelligence Applications and Innovations*.
- Pedrazzini, E., **Zangirolami, V.**, Migliorati, S., and Borrotti, M. (2026). *Flexible Beta: A novel policy distribution for proximal policy optimization.* *Statistical Science: From Theory to Applied Research IV*, accepted (to appear).
- Zanotti, M., **Zangirolami, V.**, and Pavesi, F. (2026). *An evaluation of ensemble strategies for time-series anomaly detection.* *Statistical Science: From Theory to Applied Research IV*, accepted (to appear).
- Pavesi, F., Zanotti, M., and **Zangirolami, V.** (2026). *A characterization of Gaussian processes over flat tori.* *Statistical Science: From Theory to Applied Research IV*, accepted (to appear).

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## Invited Talks

- **Zangirolami, V.**, Borrotti, M., and Candelieri, A. *Conformal prediction for safe decision-making*. **HiTEc Meeting & Workshop on Complex Data in Econometrics and Statistics**, Limassol, Cyprus, July 8–9, 2025. **Session chair**.
- **Zangirolami, V.**, Borrotti, M., and Candelieri, A. *Using conformal prediction for Gaussian process regression in dynamic systems*. **Stataik 2025**, Milan, Italy, June 13–14, 2025.
- **Zangirolami, V.** *Enhancing data efficiency in online deep reinforcement learning under partial observability*. **18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics)**, London, UK, Dec 2024.

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## Contributed Talks

- **Zangirolami, V.**, Pavesi, F., and Zanotti, M. (2026). *Safe exogenous-state reinforcement learning for water tank systems*. **Joint Meeting SIS-FENStatS 2026**, Rome, Italy, June 22–25, 2026.

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## Organization of Scientific Events

- Serving as a **Student Volunteer Organiser** for the NeurIPS 2026 Workshop *Foundations of LLM Post-Training in Changing Environments*, organized by Dr. F. Quinzan, Prof. C. Shi, Dr. G. Peyré, Prof. P. Rebeschini, Prof. F. Liu.

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## Teaching Experience (teaching's evaluations are attached)

### Teaching Assistant

*Statistics course, Bachelor's Degree*

AY 2025 – 2026

*Bocconi University, Italy*

- Supported teaching activities for the undergraduate statistics course in Italian and English.

### Tutor, Machine Learning and AI Course

*Scientific Degree Plan (PLS)*

Dec 2025 – May 2026

*University of Milano-Bicocca, Italy*

- Organized and delivered a machine learning and artificial intelligence crash course which includes also practical session testing AI models using Hugging Face.
- Language of instruction: Italian.

### Tutor

*Scientific Degree Plan (PLS)*

Dec 2025 – May 2026

*University of Milano-Bicocca, Italy*

- Prepared materials for laboratory activities and supported student projects on classification models in R.
- Language of instruction: Italian.

### Adjunct Lecturer, Business Statistics

*Bachelor's Degree in Economics and Management*

AY 2024 – 2025 / 2025 – 2026

*University of Milan, Italy*

- Delivered lectures in Italian using R programming, application of linear models, decision trees, clustering, and time series analysis.
- Language of instruction: Italian.
- Course materials available at [https://github.com/ValentinaZangirolami/Statistica\\_decisioni\\_aziendali](https://github.com/ValentinaZangirolami/Statistica_decisioni_aziendali).

### Adjunct Lecturer, Statistical Learning

*Master's Degree in Artificial Intelligence for Science and Technology*

AY 2024 – 2025

*University of Milano-Bicocca, Italy*

- Delivered lectures in English on statistical learning methods in R, including practical exercises on decision trees.
- Language of instruction: English.

- Course materials available at <https://github.com/ValentinaZangirolami/Statistical-Learning>.

**Tutor** Sep 2024  
*Stempathy Project* IISS “M.K. GANDHI”, Italy

- Prepared materials for laboratory activities and led lab sessions on regression models in R.
- Language of instruction: Italian.

**Tutor** June 2024 – July 2024  
*Summer School PNRR: Laboratory of Statistics* University of Milano-Bicocca, Italy

- Contribution to the statistics laboratory activities, supporting students in data analysis of the mosquito distribution globally.
- Language of instruction: Italian.

**Tutor, Statistical Learning** AY 2023 – 2024  
*Master’s Degree in Artificial Intelligence for Science and Technology* University of Milano-Bicocca, Italy

- Delivered tutorials in English using R programming and tidyverse workflows.
- Introduced GitHub-based project workflows; example final project: data analysis and statistical learning models for churn prediction.
- Led a mock final exam reviewing key theoretical topics in statistical learning.
- Language of instruction: English.

**Tutor** Feb 2024 – May 2024  
*Scientific Degree Plan (PLS)* University of Milano-Bicocca, Italy

- Prepared materials for laboratory activities and supported student projects on regression models in R.
- Language of instruction: Italian.

**Adjunct Lecturer, Statistical Modelling** AY 2023 – 2024  
*Bachelor’s Degree in Artificial Intelligence* University of Milano-Bicocca, Italy

- Delivered practical sessions in English on simple linear regression, multiple linear regression, and generalized linear models, with applications in R, and written exercises on hypothesis testing.
- Language of instruction: English.

**Tutor, Statistical Learning** AY 2022 – 2023  
*Master’s Degree in Artificial Intelligence for Science and Technology* University of Milano-Bicocca, Italy

- Delivered tutorials in English on introductory R programming, data analysis, and linear regression.
- Led a mock final exam reviewing key theoretical topics in statistical learning.
- Language of instruction: English.

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## Work Experience

**Data Science Analyst** July 2022 – Oct 2022  
*Accenture (Media & Communications – Telecommunications)* Milan, Italy

- Conducted data quality assessment, data analysis, feature selection, and feature engineering.
- Worked with AWS S3 and EMR, DBeaver/Apache Hive, and PySpark.

**Data Engineering Analyst** Jan 2022 – June 2022  
*Accenture (Banking & Finance – ESG)* Milan, Italy

- Developed Power BI dashboards and supported project management activities.
- Gained experience in business model strategy and business process design.

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## Grants

### AIRR Gateway Project

May 2026 - Aug 2026

- Member of the research team for the grant project *AIRR Gateway*, entitled “Reasoning-Enhanced Multimodal AI-Generated Content Detection”. PI: Prof. Giulia Livieri, Co-PI: Prof. Chengchun Shi

### AIRR Gateway Project

Feb 2026 - May 2026

- Member of the research team for the grant project *AIRR Gateway*, entitled “Lowering the Computational Barrier: Resource- and Energy-Efficient Reinforcement Learning for LLM Reasoning”. PI: Prof. Chengchun Shi, Co-PI: Prof. Giulia Livieri

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## Research Awards

### Workshops (Selected/Invited Participation)

#### Sanya Workshop on Causality and Machine Learning

Jan 12–16, 2026

*Tsinghua University – Tsinghua Sanya International Mathematics Forum (TSIMF)*

*Sanya, China*

- Invited participant.

#### The Data Research Camp

July 15–18, 2024

*Department of Statistics, Computer Science, Applications, University of Florence*

*Florence, Italy*

- Selected participant in a workshop involving group activities focused on developing new models for clinical and protein data.
- Received a travel award covering accommodation.

### Data Competitions

#### Stats Under the Stars 6

June 20–21, 2023

*Italian Statistical Society (SIS)*

*Ancona, Italy*

- Awarded second place for Best Objective Prediction.
- Challenge topic: estimation of portfolio weights for financial assets using real data provided by Flow Traders.

#### Lab Data Challenges: Electricity Load Forecast Challenge

May 2021

*University of Milano-Bicocca*

*Milan, Italy*

- Awarded first place for Best Prediction.
- Challenge topic: long-term electricity load forecasting using real data provided by Repower.

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## Referee service

- Journal of Statistical Computation and Simulation

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## Thesis Supervision

- **Co-supervisor** of master’s students working on reinforcement learning and conformal prediction topics, including: Giulia De Innocentiis (*Conformal Prediction for Neural Fitted Q-iteration*), Giorgia Faccaroni (*Conformal Prediction for Fitted Q-iteration*), Antonio Monforte (*Proximal Policy Optimization for Autonomous Driving*), and Roberto Magnacavallo (*Conformal Prediction for Superconductivity Data*).

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## Other Courses (Extra-Curricular Professional Courses)

### **Natural Language Processing Specialization**

Sep 2022

*DeepLearning.AI (Coursera)*

- Certification available upon request.

### **AI for Medicine Specialization**

Sep 2022

*DeepLearning.AI (Coursera)*

- Certification available upon request.

### **Data Engineering and Machine Learning Using Spark**

Aug 2022

*IBM (Coursera)*

- Certification available upon request.

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## Coding Project

### **Multi-Agent Deep Recurrent Q-Network | Python**

Dec 2021 – Jan 2022

- Developed a multi-agent deep recurrent Q-network for self-driving car applications.

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## Skills

**Programming Languages:** R (advanced), Python (advanced), SQL (intermediate), C/C++ (basic)

**Markup Languages:** LaTeX (advanced), HTML (basic)

**Data Visualization:** Power BI, Tableau

**Other Statistical Software:** KNIME, Gretl

**Developer Tools:** Jupyter Notebooks, Git, AWS (S3 and EMR), IBM Cloud (Watson Machine Learning)

**Operating Systems:** Windows, macOS, Linux (basic)

**Python Libraries:** NumPy, pandas, scikit-learn, TensorFlow, PyTorch, Keras, Matplotlib, Seaborn, OpenCV, PySpark, OpenAI Gym

**Languages:** Italian (native), English (fluent)

July 2026

**Valentina Zangirolami**